List R5b — Trade credit and working capital management

Worked solutions

Finanças II — Corporativas (MPE Insper)

Statements are in `Listas/lista-r5b-trade-credit-working-capital.md`; each exercise is cited by number and printed page and is not restated. Editions are the local copies: Berk = Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2nd ed. (2011); Ross = Ross, Westerfield, Jordan & Lamb, *Fundamentos de Administração Financeira*, 9th ed. (2013); Higgins = Koski, Higgins & Mitton, *Analysis for Financial Management*, 13th ed. (2023); Grinblatt = Grinblatt & Titman, *Financial Markets and Corporate Strategy*, 2nd ed. (2002). Ross prints thousands with a period ($89.804); below they are written 89,804. Money is in the units of each statement.

Conventions used throughout. A year has 365 days (all four books use 365). Days ratios use the year-flow divided by 365: inventory and payables days on cost of goods sold (COGS), receivables days on sales. The effective annual rate (EAR) of a rate r earned over n days is $(1+r)^{365/n}-1$, i.e. compounding over $365/n$ periods (Berk Eq. 5.1; Ross, p. 709).

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1. Westerly Industries: cash conversion cycle — Berk 19.3, p. 586

What is asked. Compute the cash conversion cycle (CCC): the number of days between paying suppliers for inventory and collecting cash from customers, from one year of sales and COGS and the year-end balances of receivables, inventory and payables.

Approach. The CCC is the sum of inventory days and receivables days (the operating cycle) minus payables days (Berk, §19.1). Each "days" figure converts a balance into the number of days of the matching flow it represents: inventory and payables are carried at cost, so they are divided by daily COGS; receivables are carried at selling price, so they are divided by daily sales.

Notation.

- S, C: annual sales and COGS (\$); AR, INV, AP: receivables, inventory, payables (\$).
- D_{INV}, D_{AR}, D_{AP}: inventory, receivables and payables days (days, ≥ 0).
- $OC = D_{INV} + D_{AR}$: operating cycle; $CCC = OC - D_{AP}$ (days, any sign).

Assumption. The balances are representative of the year (no averages are given).

Step 1 — Daily flows. Divide each annual flow by 365 to get the denominators of the days ratios:

$$C/365 = (80{,}000)/365 = 219.18,$$
$$S/365 = (100{,}000)/365 = 273.97.$$

Step 2 — The three days ratios.

$$D_{INV} = \frac{INV}{C/365} = \frac{(15{,}000)}{(80{,}000/365)} = 68.44 \text{ days}, \tag{1}$$

$$D_{AR} = \frac{AR}{S/365} = \frac{(30{,}000)}{(100{,}000/365)} = 109.50 \text{ days}, \tag{2}$$

$$D_{AP} = \frac{AP}{C/365} = \frac{(40{,}000)}{(80{,}000/365)} = 182.50 \text{ days}. \tag{3}$$

Step 3 — Cash conversion cycle. Add inventory and receivables days, then subtract payables days:

$$\begin{aligned} CCC &= D_{INV} + D_{AR} - D_{AP} \\ &= (68.44) + (109.50) - (182.50) \\ &= (177.94) - (182.50) \\ &= -4.56 \text{ days}. \end{aligned} \tag{4}$$

Answer. $CCC = -4.56$ days (operating cycle 177.94 days, payables 182.50 days).

Intuition. A negative CCC means suppliers wait longer to be paid (half a year) than it takes Westerly to turn inventory into cash; suppliers finance the whole operating cycle, and a little more. The firm needs no cash of its own to carry working capital — the opposite of the usual case. Whether this is good depends on why: if Westerly is forgoing early-payment discounts or stretching beyond the due date, the "free" financing may be very expensive (Exercises 7 and 13).

Common mistakes.

- Dividing payables or inventory by sales instead of COGS: it mixes cost-based balances with a price-based flow and understates both ratios.
- Reporting the operating cycle (177.94 days) as the answer, i.e. forgetting to net out payables.
- Treating the negative sign as an arithmetic error; it is a legitimate outcome.

Ref: Berk Ch. 19, §19.1 "Overview of Working Capital", The Cash Cycle, p. 567. No printed answer.

2. Mediadora: operating and cash cycles — Ross 18.6, p. 654

What is asked. Compute the operating cycle (days from buying inventory to collecting the sale) and the cash cycle (days from paying suppliers to collecting the sale) from beginning and ending balances, credit sales and COGS, and interpret them.

Approach. Ross (§18.2) computes each period from a turnover ratio built on the *average* of beginning and ending balances: inventory and payables turnover on COGS, receivables turnover on credit sales; days = 365/turnover. Operating cycle = inventory period + receivables period; cash cycle = operating cycle − payables period.

Notation.

- $\overline{INV}$, $\overline{AR}$, $\overline{AP}$: average inventory, receivables, payables ($).
- T_{INV}, T_{AR}, T_{AP}: turnovers (times per year); D_{INV}, D_{AR}, D_{AP}: periods (days).
- $S_c = 89{,}804$: credit sales; $C = 56{,}384$: COGS ($).

Step 1 — Average balances.

$$\begin{aligned}\overline{INV} &= (9{,}780 + 11{,}380)/2 = 10{,}580,\\ \overline{AR} &= (4{,}108 + 4{,}938)/2 = 4{,}523,\\ \overline{AP} &= (7{,}636 + 7{,}927)/2 = 7{,}781.5.\end{aligned}$$

Step 2 — Turnovers.

$$\begin{aligned}T_{INV} &= C/\overline{INV} = (56{,}384)/(10{,}580) = 5.3293,\\ T_{AR} &= S_c/\overline{AR} = (89{,}804)/(4{,}523) = 19.8550,\\ T_{AP} &= C/\overline{AP} = (56{,}384)/(7{,}781.5) = 7.2459.\end{aligned}$$

Step 3 — Periods in days.

$$D_{INV} = 365/T_{INV} = 365/(5.3293) = 68.49 \text{ days}, \tag{5}$$
$$D_{AR} = 365/T_{AR} = 365/(19.8550) = 18.38 \text{ days}, \tag{6}$$
$$D_{AP} = 365/T_{AP} = 365/(7.2459) = 50.37 \text{ days}. \tag{7}$$

Step 4 — The two cycles.

$$OC = D_{INV} + D_{AR} = (68.49) + (18.38) = 86.87 \text{ days}, \tag{8}$$
$$CC = OC - D_{AP} = (86.87) - (50.37) = 36.50 \text{ days}. \tag{9}$$

Answer. Operating cycle = 86.87 days; cash cycle = 36.50 days.

Interpretation. Goods sit in inventory for about 68 days and customers take about 18 days to pay, so it takes 87 days to turn a purchase into cash. Suppliers finance 50 of those days. For the remaining 36.5 days the firm has paid out cash that has not yet come back, and that gap must be financed (cash, a bank line or equity). Most of the cycle is inventory, so inventory management is where days can be cut; receivables are already quick.

Intuition. The cash cycle measures how long the firm's own money is tied up per unit of activity. Multiplied by daily COGS (56,384/365 ≈ 154 per day) it gives an order of magnitude for the working capital that must be funded; any day saved frees that much cash.

Common mistakes.

- Using ending instead of average balances; Ross uses averages when both are given.
- Building receivables turnover on COGS or payables turnover on sales.
- Interpreting the cash cycle as a profitability measure; it is a financing-need measure.

Ref: Ross Cap. 18, §18.2 "O ciclo operacional e o ciclo financeiro", pp. 627–631 (worked example in the chapter review, 18.1, pp. 651–652). No printed answer.

3. Cash discounts on a 400-unit order — Ross 20.1, p. 733

What is asked. (a) How long until the bill is due and how much is paid on the due date. (b) The size of the discount, the deadline to get it, and the discounted payment. (c) How much interest is implicitly paid by skipping the discount, and for how many days of credit.

Approach. The terms "1/10, net 30" mean: pay the full invoice by day 30, or pay 1% less by day 10. Skipping the discount is a loan from the supplier of the *discounted* amount for the days between the discount deadline and the due date; the forgone discount is the interest (Ross, §20.2, "Custo de crédito").

Notation.

- F: invoice amount (\$); $d = 0.01$: discount rate; $t_d = 10$, $t_n = 30$: discount and net days.
- r: interest rate for the $t_n - t_d$ days of credit; EAR: its effective annual equivalent.

Step 1 — Invoice amount. $F = (400)(125) = 50{,}000$.

Step 2 — (a) Full credit period. The bill is due on day $t_n = 30$; paying then costs the full $F = \$50{,}000$.

Answer (a). 30 days; pay \$50,000.

Step 3 — (b) Discount.

$$dF = (0.01)(50{,}000) = 500,$$
$$F - dF = (50{,}000) - (500) = 49{,}500.$$

Answer (b). A 1% discount (\$500) for paying within 10 days; pay \$49,500.

Step 4 — (c) Implicit interest and days of credit. Not taking the discount means keeping \$49,500 for $30 - 10 = 20$ extra days and paying \$500 more. The rate for those 20 days and its annual equivalent are

$$r = \frac{dF}{F - dF} = \frac{(500)}{(49{,}500)} = 1.0101\%, \tag{10}$$
$$\text{EAR} = (1 + r)^{365/20} - 1 = (1.010101)^{18.25} - 1 = 20.13\%. \tag{11}$$

Answer (c). Implicit interest of \$500 for 20 days of credit (1.0101% per 20 days, an EAR of 20.13%).

Intuition. The discount looks tiny, but it is earned for paying only 20 days early; repeated 18 times a year it compounds to about 20%. A buyer that can borrow below 20% should borrow and take the discount; a buyer whose marginal funding costs more should pay on day 30 — never on day 15, which gives up the discount without using the credit.

Common mistakes.

- Computing the rate on the full invoice ($500/50{,}000 = 1\%$): the amount borrowed is the discounted price.
- Counting 30 days of credit instead of 20; the first 10 days are free under both choices.
- Annualising by simple interest ($1.0101\% \times 18.25 = 18.43\%$), which understates the cost.

Ref: Ross Cap. 20, §20.2 "Condições de venda", "Custo de crédito", p. 709. No printed answer.

4. Kyoto Boy: collection period with a discount — Ross 20.3, p. 733

What is asked. (a) The average collection period (PMR) when 65% of customers pay on day 10 (taking the 2% discount) and the rest pay on day 30. (b) The average receivables balance for 1,300 sales a month at \$1,700 each.

Approach. The PMR is the payment day averaged over customers, weighted by the share that pays on each day (Ross, "O desconto financeiro e o PMR", p. 710). Receivables then follow from $AR =$ average daily sales $\times\, PMR$ (Ross, §20.1, Eq. for "Contas a receber").

Notation.

- $w = 0.65$: share of customers taking the discount; PMR: average collection period (days).
- S_m, S_a: monthly and annual sales (\$, before discounts); AR: average receivables (\$).

Assumption. Customers pay exactly on day 10 or day 30; sales are measured before discounts (as in Ross's own example, p. 710); daily sales = annual sales/365.

Step 1 — (a) Weighted collection period.

$$\begin{aligned} PMR &= w \cdot 10 + (1-w) \cdot 30 \\ &= (0.65)(10) + (0.35)(30) \\ &= 6.5 + 10.5 \\ &= 17 \text{ days.} \end{aligned} \tag{12}$$

Answer (a). $PMR = 17$ days.

Step 2 — (b) Annual and daily sales.

$$\begin{aligned} S_m &= (1{,}300)(1{,}700) = 2{,}210{,}000, \\ S_a &= 12\,(2{,}210{,}000) = 26{,}520{,}000, \\ S_a/365 &= (26{,}520{,}000)/365 = 72{,}657.53 \text{ per day.} \end{aligned}$$

Step 3 — (b) Receivables.

$$AR = (S_a/365) \times PMR = (72{,}657.53)(17) = 1{,}235{,}178.08. \tag{13}$$

Answer (b). Average receivables ≈ \$1,235,178.

Intuition. The discount pulls two-thirds of the customers forward by 20 days, so the firm carries only 17 days of sales instead of 30. The price is the 2% given away on 65% of sales; whether it pays depends on the value of the 13 days of financing released, the subject of Exercise 3 seen from the seller's side.

Common mistakes.

- Averaging the days without weights ($(10+30)/2 = 20$).
- Multiplying the PMR by *monthly* sales; with a 30-day month the result would be $2{,}210{,}000 \times 17/30 = 1{,}252{,}333.33$, a different (and inconsistent) day count.
- Netting the discount out of sales without saying so; it changes the answer by about 1%.

Ref: Ross Cap. 20, §20.1 "Crédito e contas a receber" and §20.2, "O desconto financeiro e o PMR", p. 710. No printed answer.

5. Happy Valley Homecare: inventory days — Berk 19.17, p. 589

What is asked. (a) How many days of COGS HVHS holds in inventory. (b) How much less inventory it would carry at the industry's 73 days.

Approach. Inventory days = inventory/(COGS/365) (Berk, §19.1 and §19.5). Inverting the ratio, the inventory a firm must carry at a target number of days is target days × daily COGS; the difference with the current balance is cash released once.

Notation.

- $C = 8{,}000{,}000$: COGS; $\overline{INV} = 2{,}000{,}000$: average inventory ($).
- D_{INV}: inventory days; $D^* = 73$: industry days; INV^*: inventory at D^* ($).

Step 1 — (a) Daily COGS and inventory days.

$$\begin{aligned} C/365 &= (8{,}000{,}000)/365 = 21{,}917.81, \\ D_{INV} &= \overline{INV}/(C/365) = (2{,}000{,}000)/(21{,}917.81) = 91.25 \text{ days}. \end{aligned} \tag{14}$$

Answer (a). $D_{INV} = 91.25$ days.

Step 2 — (b) Inventory at the industry average.

$$\begin{aligned} INV^* &= D^*\,(C/365) = (73)(21{,}917.81) = 1{,}600{,}000, \\ \overline{INV} - INV^* &= (2{,}000{,}000) - (1{,}600{,}000) = 400{,}000. \end{aligned} \tag{15}$$

Check: $(91.25 - 73) \times 21{,}917.81 = (18.25)(21{,}917.81) = 400{,}000$.

Answer (b). Inventory investment falls by $400,000.

Intuition. Each day of inventory ties up one day of COGS. Cutting 18.25 days frees $400,000 once (a one-time cash inflow), plus the carrying cost on it every year after. Sales ($20 million) play no role: inventory is at cost.

Common mistakes.

- Using sales instead of COGS ($2{,}000{,}000/(20{,}000{,}000/365) = 36.5$ days), which makes HVHS look far better than the industry.
- Reporting the saving as annual; it is a one-time release of capital.

Ref: Berk Ch. 19, §19.1, p. 567; §19.5 "Inventory Management", p. 580. No printed answer.

6. The Greek Connection: NWC and the CCC — Berk 19.6, p. 587

What is asked. (a) Net working capital from the balance sheet. (b) The cash conversion cycle. (c) The CCC if receivables days had been the industry's 30 days.

Approach. Berk defines net working capital as current assets minus current liabilities (§19.1); the CCC is built as in Exercise 1. Part (c) changes only the receivables component, so the CCC moves one-for-one with receivables days.

Notation.

- CA, CL: current assets and liabilities ($ thousand); $NWC = CA - CL$.
- $S = 32{,}000$, $C = 20{,}000$: sales and COGS ($ thousand); days ratios as in Exercise 1.

Step 1 — (a) Net working capital.

$$\begin{aligned} CA &= 2{,}000 + 3{,}950 + 1{,}300 = 7{,}250, \\ CL &= 1{,}500 + 1{,}000 + 1{,}220 = 3{,}720, \\ NWC &= CA - CL = (7{,}250) - (3{,}720) = 3{,}530. \end{aligned} \tag{16}$$

Answer (a). $NWC = \$3{,}530$ thousand ($3.53 million).
Remark: the *operating* part, which excludes cash and interest-bearing notes, is $AR + INV - AP -$ Accruals $= 3{,}950 + 1{,}300 - 1{,}500 - 1{,}220 = 2{,}530$ thousand.

Step 2 — (b) Daily flows and days ratios.

$$\begin{aligned} S/365 = (32{,}000)/365 = 87.671, \qquad C/365 = (20{,}000)/365 = 54.795, \\ D_{AR} = (3{,}950)/(87.671) = 45.05 \text{ days}, \\ D_{INV} = (1{,}300)/(54.795) = 23.73 \text{ days}, \\ D_{AP} = (1{,}500)/(54.795) = 27.38 \text{ days}. \end{aligned}$$

Step 3 — (b) Cash conversion cycle.

$$CCC = D_{INV} + D_{AR} - D_{AP} = (23.73) + (45.05) - (27.38) = 41.40 \text{ days}. \tag{17}$$

Answer (b). $CCC = 41.40$ days.

Step 4 — (c) Receivables at 30 days. Replace D_{AR} by 30 in (17):

$$\begin{aligned} CCC' = (23.73) + (30) - (27.38) = 26.35 \text{ days}, \\ CCC - CCC' = (41.40) - (26.35) = 15.05 \text{ days}. \end{aligned} \tag{18}$$

The receivables balance consistent with 30 days and the cash it would release are

$$\begin{aligned} AR' = (30)(87.671) = 2{,}630.14, \\ AR - AR' = (3{,}950) - (2{,}630.14) = 1{,}319.86 \text{ thousand}. \end{aligned}$$

Answer (c). $CCC' = 26.35$ days (15.05 days shorter), releasing about $1.32 million of receivables.

Intuition. Greek Connection's customers take 45 days against an industry norm of 30: the firm is lending to its customers for an extra 15 days. Matching the norm would cut the CCC by a third and release $1.32 million — more than a third of its NWC — that could repay the notes payable.

Common mistakes.

- Computing receivables days on COGS ($3{,}950/54.795 = 72$ days).
- Leaving notes payable out of current liabilities in (a); NWC in Berk's definition includes them.
- In (c), recomputing all three components; only receivables days change.

Ref: Berk Ch. 19, §19.1 "Overview of Working Capital", pp. 566–567; §19.3 "Receivables Management", p. 576. No printed answer.

7. Terms 3/15, net 40 — Berk 19.14, p. 588

What is asked. The effective annual cost of skipping a 3% discount (a) when paying on the due date, day 40, and (b) when stretching payment to day 50.

Approach. Skipping the discount borrows the discounted price, 97% of the invoice, from day 15 until the payment day, at a cost of 3% of the invoice. The rate per period is 3/97; the period is the number of days of credit actually used, and the EAR compounds that period over a 365-day year (Berk, Example 19.3 and Example 19.7, "stretching the accounts payable").

Notation.

- $d = 0.03$: discount; $t_d = 15$: discount deadline; t: actual payment day.
- $r = d/(1-d)$: rate per credit period; $n = t - t_d$: days of credit (days).

Step 1 — Rate per credit period. On a \$100 invoice, pay \$97 on day 15 or \$100 later: \$3 of interest on \$97 borrowed,

$$r = \frac{d}{1-d} = \frac{(0.03)}{(0.97)} = 3.0928\%. \tag{19}$$

Step 2 — (a) Pay on day 40. Credit period $n = 40 - 15 = 25$ days; there are $365/25 = 14.6$ such periods per year:

$$\begin{aligned} \text{EAR}_{40} &= (1+r)^{365/25} - 1 \\ &= (1.030928)^{14.6} - 1 \\ &= 56.00\%. \end{aligned} \tag{20}$$

Answer (a). Effective annual cost = 56.00%.

Step 3 — (b) Pay on day 50. Now $n = 50 - 15 = 35$ days and $365/35 = 10.4286$ periods:

$$\begin{aligned} \text{EAR}_{50} &= (1.030928)^{10.4286} - 1 \\ &= 37.39\%. \end{aligned} \tag{21}$$

Answer (b). Effective annual cost = 37.39%.

Intuition. The interest charge is fixed (\$3 per \$97), so the longer the firm keeps the money, the lower the rate per day: stretching from 40 to 50 days cuts the cost from 56% to 37%. But both are far above normal bank rates, so the first-best is usually to borrow from the bank and take the discount. Stretching has hidden costs the formula does not see — late-payment penalties, a damaged credit rating, suppliers that refuse future credit.

Common mistakes.

- Using 40 or 50 days as the period instead of the days beyond the discount date (25, 35).
- Using 3/100 instead of 3/97.
- Reporting the APR $3.0928\% \times 14.6 = 45.15\%$ as the cost; Berk's "Common Mistake" box (p. 572) warns that the EAR is the right comparison.

Ref: Berk Ch. 19, §19.2 "Trade Credit", Cost of Trade Credit, pp. 571–572 (Example 19.3); §19.4 "Payables Management", Stretching Accounts Payable, p. 579 (Example 19.7). No printed answer.

8. Toyota vs. Apple: operating and cash conversion cycles — Higgins 2.13, p. 75

What is asked. (a) The operating cycle of each firm (inventory days + collection period) and what it means. (b) The cash conversion cycle (operating cycle − payables period) and what it means. (c) The pros and cons of Apple's way of running operations relative to Toyota's.

Approach. Higgins (Ch. 2, "Operating and Cash Conversion Cycles") defines days inventory outstanding and the payables period on COGS, the collection period on sales, all with 365 days. Because each firm's ratios are in its own currency per day, the yen/dollar difference cancels inside each ratio and the days are directly comparable.

Notation.

- D_{INV}, D_{AR}, D_{AP}: days inventory outstanding, collection period, payables period (days).
- $OC = D_{INV} + D_{AR}$; $CCC = OC - D_{AP}$ (days). Figures in millions (¥ or \$).

Step 1 — Daily flows.

$$\text{Toyota: } C/365 = (24{,}522{,}364)/365 = 67{,}184.56, \qquad S/365 = (29{,}929{,}992)/365 = 82{,}000.0,$$
$$\text{Apple: } C/365 = (169{,}559)/365 = 464.55, \qquad S/365 = (274{,}515)/365 = 752.10.$$

Step 2 — (a) Inventory days and collection period.

$$D^T_{INV} = (2{,}434{,}918)/(67{,}184.56) = 36.24, \qquad D^T_{AR} = (8{,}709{,}065)/(82{,}000.0) = 106.21,$$
$$D^A_{INV} = (4{,}061)/(464.55) = 8.74, \qquad D^A_{AR} = (16{,}120)/(752.10) = 21.43.$$

Step 3 — (a) Operating cycles.

$$OC^T = (36.24) + (106.21) = 142.45 \text{ days}, \tag{22}$$
$$OC^A = (8.74) + (21.43) = 30.18 \text{ days}. \tag{23}$$

Answer (a). Toyota 142.45 days; Apple 30.18 days. On average Toyota needs about 142 days from acquiring inventory to collecting the customer's cash, Apple only 30. Toyota's cycle is dominated by receivables: 106 days, which reflects dealer financing and Toyota's captive finance arm (customer loans and leases sit in receivables).

Step 4 — (b) Payables periods.

$$D^T_{AP} = (2{,}434{,}180)/(67{,}184.56) = 36.23 \text{ days},$$
$$D^A_{AP} = (42{,}296)/(464.55) = 91.05 \text{ days}.$$

Step 5 — (b) Cash conversion cycles.

$$CCC^T = (142.45) - (36.23) = 106.22 \text{ days}, \tag{24}$$
$$CCC^A = (30.18) - (91.05) = -60.87 \text{ days}. \tag{25}$$

Answer (b). Toyota 106.22 days; Apple −60.87 days. Toyota pays for its inputs about 106 days before it collects, so it must finance that gap; Apple collects from customers about 61 days *before* it pays its suppliers, so its suppliers finance its operations and more. In balance sheet terms, Apple's trade working capital $AR + INV - AP = 16{,}120 + 4{,}061 - 42{,}296 = -22{,}115$ million is a source of funds, while Toyota's, $8{,}709{,}065 + 2{,}434{,}918 - 2{,}434{,}180 = 8{,}709{,}803$ million yen, is a large use.

Step 6 — (c) Advantages and disadvantages of Apple's model.

- *Advantages.* (i) No need to finance working capital; growth generates cash instead of consuming it, because each extra dollar of sales adds more payables than receivables and inventory. (ii) Little inventory means little obsolescence risk in a fast-changing product line. (iii) Bargaining power over suppliers and outsourced assembly lets Apple push the holding of inventory and the waiting for payment upstream.
- *Disadvantages and risks.* (i) Thin inventory risks stock-outs and lost sales when demand spikes or a supply chain is disrupted. (ii) Demanding quick payment can drive customers (carriers, retailers) away. (iii) Paying slowly can strain or alienate suppliers, raise the prices they charge, or create fragility if a supplier fails. (iv) The model depends on market power; a firm without Apple's brand cannot copy it.
- *Why Toyota differs.* Selling cars on credit (dealer floor-plan and consumer financing) is part of Toyota's product offer; the long receivables period is a business choice funded by its financial-services arm, not a collection failure.

Intuition. The CCC tells you who finances whom along the supply chain. A negative cycle is a quasi-free loan from suppliers, available only to firms with market power and fast inventory turnover.

Common mistakes.

- Converting yen to dollars before computing days; it is unnecessary because each ratio is within one currency.
- Basing the payables period on sales instead of COGS.
- Calling Apple's negative CCC "risk-free"; it comes with supplier and stock-out risks.

Book answer. Higgins p. 404: DIO 36.2 and 8.7 days; collection period 106.2 and 21.4 days; operating cycle 142.4 and 30.1 days; payables 36.2 and 91.0 days; CCC 106.2 and −60.9 days; (c) the same advantages and risks. Matches the book's answer; the book adds rounded components (e.g. $36.2 + 106.2 = 142.4$, $8.7 + 21.4 = 30.1$), which explains the differences in the second decimal.

Ref: Higgins Ch. 2, "Operating and Cash Conversion Cycles", p. 47.

9. Hand-to-Mouth: forgone discount vs. two bank loans — Berk 20.8, p. 613

What is asked. Which of three ways to get \$10,000 of usable funds for about 30 days is cheapest: (A) skipping a 2/10, net 30 discount; (B) a 12% APR loan with a 5% non-interest compensating balance and a \$100 fee; (C) a 15% APR loan with a 1% origination fee.

Approach. Compare effective annual rates computed on the cash the firm can actually use. A fee paid up front and a compensating balance both reduce usable proceeds while interest is charged on the face value, so the loan must be grossed up until usable funds equal \$10,000 (Berk, §20.3, "Common Loan Stipulations and Fees"). The trade-credit cost is as in Exercise 7.

Notation.

- L: face value borrowed (\$); i: interest for the 30-day loan (\$); r: 30-day rate on usable funds.
- $\text{EAR} = (1 + r)^{365/n} - 1$ with n the length of the credit period (days).

Assumption. A 30-day loan is one month: interest $= (\text{APR}/12)\, L$, paid at maturity. The compensating balance is applied to the repayment at maturity (Berk's Timmons example, p. 603). All three rates are annualised with 365 days, $365/n$ periods, so they are comparable.

Step 1 — Alternative A: forgo the discount. Paying \$98 on day 10 or \$100 on day 30 means borrowing \$98 for $n = 20$ days:

$$\begin{aligned} r_A &= (2)/(98) = 2.0408\%, \\ \text{EAR}_A &= (1.020408)^{365/20} - 1 = (1.020408)^{18.25} - 1 = 44.59\%. \end{aligned} \tag{26}$$

Step 2 — Alternative B: gross up the face value. Usable proceeds $= L - 0.05L - 100 = 10{,}000$, so

$$\begin{aligned} 0.95\, L &= 10{,}000 + 100 = 10{,}100, \\ L &= (10{,}100)/(0.95) = 10{,}631.58. \end{aligned}$$

The compensating balance is $0.05\, L = 531.58$. Interest and net repayment at day 30:

$$\begin{aligned} i_B &= (0.12/12)(10{,}631.58) = 106.32, \\ \text{Net repayment} &= L + i_B - 0.05L = (10{,}631.58) + (106.32) - (531.58) = 10{,}206.32, \\ r_B &= (10{,}206.32)/(10{,}000) - 1 = 2.0632\%, \\ \text{EAR}_B &= (1.020632)^{365/30} - 1 = (1.020632)^{12.1667} - 1 = 28.21\%. \end{aligned} \tag{27}$$

Step 3 — Alternative C: gross up for the 1% fee. Usable proceeds $= 0.99\,L = 10{,}000$:

$$\begin{aligned} L &= (10{,}000)/(0.99) = 10{,}101.01 \quad (\text{fee } 0.01L = 101.01),\\ i_C &= (0.15/12)(10{,}101.01) = 126.26,\\ \text{Repayment} &= (10{,}101.01) + (126.26) = 10{,}227.27,\\ r_C &= (10{,}227.27)/(10{,}000) - 1 = 2.2727\%,\\ \text{EAR}_C &= (1.022727)^{12.1667} - 1 = 31.45\%. \end{aligned} \tag{28}$$

Step 4 — Ranking. $\text{EAR}_B = 28.21\% < \text{EAR}_C = 31.45\% < \text{EAR}_A = 44.59\%$. With 12 monthly periods instead of 365/30 the loan EARs are 27.77% and 30.95%; the ranking is unchanged.

Answer. Alternative B (Bank A) is cheapest, at an EAR of about 28%; forgoing the discount is by far the most expensive (44.6%).

Intuition. The stated APRs (12% vs. 15%) understate both loans' costs, because the borrower pays interest on money it cannot use (the fee, the idle balance). Bank A's lower APR survives these add-ons. Trade credit that "costs nothing" in cash outlay is the most expensive source here: 2% for only 20 days. Note also that alternative A finances 20 days, not the 30 needed — another reason to prefer a bank loan and take the discount.

Common mistakes.

- Comparing APRs (12%, 15%) directly, ignoring the fee and the compensating balance.
- Borrowing only \$10,000 in B and C: usable proceeds would then fall short of the need.
- Forgetting that the compensating balance is returned at maturity (it offsets the repayment).

Ref: Berk Ch. 20, §20.3 "Short-Term Financing with Bank Loans", Common Loan Stipulations and Fees, pp. 602–603; Ch. 19, §19.2, pp. 571–572. No printed answer.

10. Peças Aéreas: one-off vs. repeat order — Ross 20.9, pp. 733–734

What is asked. (a) Whether to grant one period of credit on a one-off order of eight engines with a 0.5% default chance. (b) The default probability that makes that NPV zero. (c) The same two questions when a customer who pays becomes a permanent, never-defaulting customer. (d) Why repeat business makes credit terms more liberal.

Approach. Ross's one-period credit model (§20.4, context block in the list): granting credit risks the variable cost now against the expected price next period. For a one-off sale, NPV $= -v + (1-\pi)P/(1+R)$ [20.8]. With repeat business a paying customer is worth a perpetuity of margins, NPV $= -v + (1-\pi)(P-v)/R$ [20.9]. Break-even probabilities set each NPV to zero.

Notation.

- $v = 1.6$: variable cost per engine; $P = 1.87$: credit price per engine (\$ million).
- $\pi = 1/200 = 0.005$: default probability; $R = 0.029$: required return per period.
- π^*: break-even default probability. The order is 8 engines.

Step 1 — (a) NPV per engine, one-off sale.

$$\begin{aligned} \text{NPV} &= -v + \frac{(1-\pi)P}{1+R}\\ &= -1.6 + \frac{(0.995)(1.87)}{1.029}\\ &= -1.6 + \frac{1.86065}{1.029}\\ &= -1.6 + 1.808212\\ &= 0.208212 \text{ per engine.} \end{aligned} \tag{29}$$

For the order: $8 \times 0.208212 = 1.665695$ (\$ million).

Answer (a). Yes: NPV ≈ \$1.67 million for the eight-engine order (\$208,212 per engine).

Step 2 — (b) Break-even default probability. Set (29) to zero and solve for π:

$$\begin{aligned} 0 &= -v + (1-\pi^*)P/(1+R) \\ (1-\pi^*)P &= v(1+R) \\ 1-\pi^* &= \frac{v(1+R)}{P} = \frac{(1.6)(1.029)}{1.87} = \frac{1.6464}{1.87} = 0.880428 \\ \pi^* &= 1 - 0.880428 = 0.119572. \end{aligned} \tag{30}$$

Answer (b). $\pi^* \approx 11.96\%$.

Step 3 — (c) Repeat business. A paying customer generates $P - v = 1.87 - 1.6 = 0.27$ per engine every period forever:

$$\begin{aligned} \text{NPV} &= -v + \frac{(1-\pi)(P-v)}{R} \\ &= -1.6 + \frac{(0.995)(0.27)}{0.029} \\ &= -1.6 + \frac{0.26865}{0.029} \\ &= -1.6 + 9.263793 \\ &= 7.663793 \text{ per engine,} \end{aligned} \tag{31}$$

i.e. $8 \times 7.663793 = 61.310345$ (\$ million) for the order. Break-even:

$$\begin{aligned} (1-\pi^*)(P-v)/R &= v \\ 1-\pi^* &= \frac{vR}{P-v} = \frac{(1.6)(0.029)}{0.27} = \frac{0.0464}{0.27} = 0.171852 \\ \pi^* &= 1 - 0.171852 = 0.828148. \end{aligned} \tag{32}$$

Answer (c). Yes: NPV ≈ \$61.31 million for the order; break-even default probability ≈ 82.81%.

Step 4 — (d) Why repeat business liberalises credit.

- In a one-off sale the firm risks v to win one payment P; the upside is a single margin.
- With repeat business, the first credit sale is a screening device: risking v once buys, with probability $1-\pi$, a customer whose value is the perpetuity $(P-v)/R$. That prize is many times P (here 9.31 vs. 1.87 per engine), so the firm tolerates a much higher default rate on new customers (82.8% vs. 12.0%).
- Defaulters reveal themselves after one period and are dropped, so the loss is capped at one unit of variable cost, while good customers keep paying. Credit is cheap information about who is a good customer.

Intuition. Granting credit to a new customer is an option-like investment: limited downside (v once), large upside (a stream of margins). The more valuable the relationship, the looser the credit standard.

Common mistakes.

- Discounting the perpetuity in (c) by $1+R$ or using P instead of $P-v$: the customer pays P but the firm spends v again on each repeat order.
- Computing the NPV with π applied to the cost as well as the price; the cost is spent for sure.
- Forgetting to multiply by the eight engines when reporting the order's NPV.

Ref: Ross Cap. 20, §20.4 "Política de crédito ótima" / "Quando o crédito deve ser concedido?" (Uma venda única; Negócios repetidos), pp. 717–718. No printed answer.

11. Lince S/A: seasonal cash budget and short-term plan — Ross 18.15, p. 657

What is asked. (a) Fill in the quarterly cash budget: net cash inflow, ending cash and the cumulative surplus over the \$30 million minimum. (b) Build the short-term financial plan in which surpluses earn 2% per quarter and deficits are borrowed at 3% per quarter, and give the net cash cost (interest paid minus investment income) for the year.

Approach. Ross's cash budget (§18.4): collections follow from the collection period, payments from the payables period, both as fractions of a 90-day quarter; net cash inflow = collections − disbursements. The short-term plan (§18.6) then invests any cash above the minimum and borrows to cover any shortfall, with interest computed on the beginning-of-quarter balances.

Notation.

- S_q: sales in quarter q; $Pur_q = 0.45\,S_{q+1}$: purchases in q (\$ million).
- Col_q, Pay_q: collections and payments to suppliers; NCF_q: net cash inflow.
- I_q: short-term investments at the end of q; $Inc_q = 0.02\,I_{q-1}$: investment income.

Assumption. A quarter has 90 days. Collection period 45 days: $45/90 = 1/2$ of a quarter's sales is collected in the quarter, the other half in the next; the \$68 million opening receivables are collected in Q1. Suppliers paid in 36 days: $36/90 = 0.4$ of a quarter's purchases is paid the next quarter, 0.6 in the same quarter. Last year's Q4 purchases were $0.45 \times 210 = 94.5$ (45% of this year's Q1 sales). The book's first row, "Meta para o saldo de caixa \$30", is read as the beginning cash balance, which the statement fixes at \$64 million (the \$30 is the minimum balance, and the part (b) table itself starts at \$64).

Step 1 — Collections.

$$\begin{aligned} Col_1 &= 68 + \tfrac{1}{2}(210) = 68 + 105 = 173, \\ Col_2 &= \tfrac{1}{2}(210) + \tfrac{1}{2}(180) = 105 + 90 = 195, \\ Col_3 &= \tfrac{1}{2}(180) + \tfrac{1}{2}(245) = 90 + 122.5 = 212.5, \\ Col_4 &= \tfrac{1}{2}(245) + \tfrac{1}{2}(280) = 122.5 + 140 = 262.5. \end{aligned}$$

Step 2 — Purchases and payments to suppliers. Purchases: $Pur_1 = 0.45(180) = 81$, $Pur_2 = 0.45(245) = 110.25$, $Pur_3 = 0.45(280) = 126$, $Pur_4 = 0.45(240) = 108$. Payments $Pay_q = 0.4\,Pur_{q-1} + 0.6\,Pur_q$:

$$\begin{aligned} Pay_1 &= 0.4(94.5) + 0.6(81) = 37.8 + 48.6 = 86.4, \\ Pay_2 &= 0.4(81) + 0.6(110.25) = 32.4 + 66.15 = 98.55, \\ Pay_3 &= 0.4(110.25) + 0.6(126) = 44.1 + 75.6 = 119.7, \\ Pay_4 &= 0.4(126) + 0.6(108) = 50.4 + 64.8 = 115.2. \end{aligned}$$

Step 3 — Disbursements and net cash inflow. Wages, taxes and other: $0.25S_q$ = 52.5, 45, 61.25, 70; interest and dividends 12 each quarter; capital spending 80 in Q2.

	Q1	Q2	Q3	Q4
Payments to suppliers	86.40	98.55	119.70	115.20
Wages, taxes, other	52.50	45.00	61.25	70.00
Interest and dividends	12.00	12.00	12.00	12.00
Capital spending	0.00	80.00	0.00	0.00
Total disbursements	150.90	235.55	192.95	197.20
Collections	173.00	195.00	212.50	262.50
Net cash inflow	22.10	−40.55	19.55	65.30

For example $NCF_2 = 195 - (98.55 + 45 + 12 + 80) = 195 - 235.55 = -40.55$.

Step 4 — (a) Cash budget. Ending cash = beginning cash + NCF_q; cumulative surplus = ending cash -30:

	Q1	Q2	Q3	Q4
Beginning cash balance	64.00	86.10	45.55	65.10
Net cash inflow	22.10	−40.55	19.55	65.30
Ending cash balance	86.10	45.55	65.10	130.40
Minimum cash balance	30.00	30.00	30.00	30.00
Cumulative surplus (deficit)	56.10	15.55	35.10	100.40

Answer (a). Ending cash 86.10, 45.55, 65.10, 130.40; cumulative surplus 56.10, 15.55, 35.10, 100.40 ($ million). There is never a deficit.

Step 5 — (b) Short-term plan. Hold exactly $30 million; invest the excess; earn 2% on the investments held at the start of each quarter. Q1: invest $64 + 22.10 - 30 = 56.10$. Q2: income $0.02(56.10) = 1.122$; the deficit $-40.55 + 1.122 = -39.428$ is covered by selling investments, so $I_2 = 56.10 - 39.428 = 16.672$. Q3: income $0.02(16.672) = 0.33344$; invest $19.55 + 0.33344 = 19.88344$; $I_3 = 36.55544$. Q4: income $0.02(36.55544) = 0.73111$; invest $65.30 + 0.73111 = 66.03111$; $I_4 = 102.58655$.

	Q1	Q2	Q3	Q4
Beginning cash balance	64.000	30.000	30.000	30.000
Net cash inflow	22.100	−40.550	19.550	65.300
New short-term investments	56.100	0.000	19.883	66.031
Income from short-term investments	0.000	1.122	0.333	0.731
Short-term investments sold	0.000	39.428	0.000	0.000
New short-term borrowing	0	0	0	0
Interest on short-term borrowing	0	0	0	0
Short-term borrowing repaid	0	0	0	0
Ending cash balance	30.000	30.000	30.000	30.000
Minimum cash balance	30.000	30.000	30.000	30.000
Cumulative surplus (deficit)	0.000	0.000	0.000	0.000
Beginning short-term investments	0.000	56.100	16.672	36.555
Ending short-term investments	56.100	16.672	36.555	102.587
Beginning short-term debt	0	0	0	0
Ending short-term debt	0	0	0	0

Net cash cost = interest paid − investment income:

$$0 - (0 + 1.122 + 0.33344 + 0.73111) = -2.18655. \tag{33}$$

Answer (b). No borrowing is ever needed; the net cash cost is $-$2.19 million, i.e. a net investment *income* of about $2.19 million for the year.

Intuition. Lince's seasonal pattern (big Q2 capital spending, strong Q4 collections) is fully covered by the opening cash cushion of $64 million: the Q2 dip is met by selling part of what was invested in Q1. The cash budget is the tool that reveals this timing; annual totals alone would hide the Q2 squeeze. With a lower opening balance the same plan would show Q2 borrowing at 3%.

Common mistakes.

- Basing purchases on the current quarter's sales instead of next quarter's, or forgetting last year's Q4 purchases (94.5) in the Q1 payment.
- Collecting 45/90 of the *previous* quarter's sales only, i.e. swapping the two halves.

- Computing investment income on end-of-quarter balances (it is earned on what was invested during the quarter, the beginning balance).

Ref: Ross Cap. 18, §18.4 "O orçamento de caixa", pp. 639–642; §18.6 "Um plano financeiro de curto prazo", pp. 649–650. No printed answer.

12. BCD Manufacturing: tax shield vs. lost supplier credit — Grinblatt 17.5, p. 619

What is asked. Whether BCD should borrow \$1 billion to buy back 40% of its stock, when the debt's tax saving is \$48 million a year but suppliers threaten to withdraw net-30 terms, which would cost an extra 2% a year on \$1.5 billion of inventory; and why.

Approach. Grinblatt & Titman's stakeholder theory of capital structure (§17.1): leverage creates tax benefits but imposes costs through non-financial stakeholders — here suppliers who fear being unpaid and react by tightening credit. Compare the two flows on the same (after-tax, annual) basis, then ask what else the suppliers' reaction signals.

Notation.

- $D = 1{,}000$: new debt; $r_D = 0.12$: interest rate; $\tau = 0.40$: marginal tax rate (\$ million).
- $INV = 1{,}500$: inventory; $c = 0.02$: extra annual financing cost if trade credit is lost.

Assumption. The \$48 million and the 2% cost are annual and recur as long as the debt is in place; the 2% is a pre-tax, tax-deductible financing cost (it replaces trade credit with costlier funds).

Step 1 — Annual tax saving.

$$\begin{aligned} \text{Interest} &= r_D D = (0.12)(1{,}000) = 120, \\ \text{Tax saving} &= \tau\, r_D D = (0.40)(120) = 48. \end{aligned} \tag{34}$$

Step 2 — Annual cost of losing trade credit.

$$\begin{aligned} \text{Pre-tax cost} &= c\, INV = (0.02)(1{,}500) = 30, \\ \text{After-tax cost} &= (1-\tau)(30) = (0.60)(30) = 18. \end{aligned} \tag{35}$$

Step 3 — Net annual effect.

$$48 - 18 = 30 \text{ per year} \quad \text{(even with no tax deduction: } 48 - 30 = 18 > 0\text{)}.$$

In present value, with perpetual debt, the tax shield is $\tau D = (0.40)(1{,}000) = 400$, while the after-tax supplier cost discounted at 12% is $18/0.12 = 150$; the net is $400 - 150 = 250$.

Step 4 — Qualitative judgement. The narrow arithmetic favours the repurchase, but the suppliers' reaction is a warning sign:

- Suppliers are a stakeholder group that bears costs in distress; if they react, customers (who worry about warranties and spare parts) and employees (who worry about their jobs) may too, and those indirect distress costs are not in the \$18 million.
- The \$48 million is not riskless: it is earned only if BCD has taxable income to shelter; higher leverage raises the chance of losses and of financial distress.
- Personal taxes on interest reduce the net tax advantage of debt (Miller), so \$48 million overstates the gain to shareholders.
- The supplier credit could perhaps be preserved by negotiation (collateral, shorter terms, a smaller repurchase).

Answer. On the numbers given, go ahead: the tax saving ($48 million) exceeds the after-tax cost of lost trade credit ($18 million) by $30 million a year ($250 million in PV for perpetual debt). But management should first weigh the indirect costs the supplier threat foreshadows; if other stakeholders react similarly, the decision can flip.

Intuition. Capital structure is not only a deal between shareholders and lenders. Firms whose suppliers, customers or employees make firm-specific investments pay for leverage through worse terms from them. The optimal debt ratio is lower when those stakeholders are important and easily spooked.

Common mistakes.

- Comparing the $48 million tax saving with the $120 million of interest: interest replaces dividends to equity; only the tax saving is a gain to the firm.
- Comparing an after-tax gain with a pre-tax cost without saying so.
- Stopping at the arithmetic and ignoring what the suppliers' reaction implies about distress costs.

Ref: Grinblatt & Titman Ch. 17, §17.1 "The Stakeholder Theory of Capital Structure", pp. 597–600. No printed answer.

13. International Motor Corporation: CCC and payables — Berk 19.15, p. 588

What is asked. (a) IMC's cash conversion cycle in 2009 and 2010, what changed, and what that implies for its cash needs. (b) Whether, with suppliers' terms of net 30, IMC manages its payables well.

Approach. The CCC as in Exercise 1, computed for each year from that year's flows and year-end balances, then decomposed to see which component moved. For (b), compare payables days with the 30 days allowed: with net terms and no discount, paying early has no benefit (Berk, §19.4, "Determining Accounts Payable Days Outstanding").

Notation.

- D^y_{INV}, D^y_{AR}, D^y_{AP}, CCC^y for year $y \in \{2009, 2010\}$ (days).
- Flows and balances in $ million.

Step 1 — Daily flows.

$$\begin{aligned} 2009: &\quad C/365 = (52{,}000)/365 = 142.47, \quad S/365 = (60{,}000)/365 = 164.38, \\ 2010: &\quad C/365 = (61{,}000)/365 = 167.12, \quad S/365 = (75{,}000)/365 = 205.48. \end{aligned}$$

Step 2 — (a) 2009.

$$\begin{aligned} D_{INV} &= (6{,}200)/(142.47) = 43.52, \quad D_{AR} = (2{,}800)/(164.38) = 17.03, \\ D_{AP} &= (3{,}600)/(142.47) = 25.27, \\ CCC^{2009} &= (43.52) + (17.03) - (25.27) = 35.28 \text{ days}. \end{aligned} \tag{36}$$

Step 3 — (a) 2010.

$$\begin{aligned} D_{INV} &= (6{,}600)/(167.12) = 39.49, \quad D_{AR} = (6{,}900)/(205.48) = 33.58, \\ D_{AP} &= (4{,}600)/(167.12) = 27.52, \\ CCC^{2010} &= (39.49) + (33.58) - (27.52) = 45.55 \text{ days}. \end{aligned} \tag{37}$$

Step 4 — (a) What changed. $CCC^{2010} - CCC^{2009} = 45.55 - 35.28 = 10.26$ days. By component: inventory -4.03 days, receivables $+16.55$ days, payables $+2.26$ days (which shortens the CCC). The deterioration is entirely receivables: had 2010 receivables days stayed at 17.03, receivables would have been $17.03 \times 205.48 = 3{,}500$ instead of 6,900, i.e. \$3,400 million less. The extra 10.26 days of the CCC, valued at 2010 daily sales, tie up about $10.26 \times 205.48 \approx$ \$2,109 million.

Answer (a). CCC = 35.28 days (2009) and 45.55 days (2010), a rise of 10.26 days driven by slower collections (receivables days 17.03 → 33.58). All else equal IMC needs more cash (on the order of \$2 billion) to finance working capital.

Step 5 — (b) Payables against net 30. Payables days were 25.27 (2009) and 27.52 (2010), both below the 30 days suppliers allow. With net-30 terms there is no discount for paying early, so every day paid before day 30 is free financing given up. Paying on day 30 in 2010 would mean payables of $30 \times 167.12 = 5{,}013.70$ instead of 4,600, i.e. \$413.70 million more interest-free credit ($30 \times 142.47 - 3{,}600 = 673.97$ in 2009).

Answer (b). No: IMC pays in about 25–28 days although it may take 30 at no cost; it could raise roughly \$414 million of free financing (2010) by paying on the due date. It has improved slightly from 2009, and it should not stretch beyond 30 days.

Intuition. IMC's working capital got worse because customers pay slower — possibly a deliberate loosening of credit to push the 25% sales growth, possibly a collection problem; the aging schedule would tell. On the payables side it leaves free money on the table. Both point to working capital as a cheap source of the cash its growth needs.

Common mistakes.

- Using one year's flows with the other year's balances.
- Advising IMC to stretch payables beyond 30 days; the costless target is exactly the due date.
- Reading the higher CCC as "more cash on hand"; it means more cash is *needed*.

Ref: Berk Ch. 19, §19.1, p. 567; §19.4 "Payables Management", pp. 578–579. No printed answer.

14. Senna S/A: switch-to-credit NPV and break-even price — Ross 20.18 (with 20.14), pp. 734–735

What is asked. Part 0 (Problem 14): whether to switch from cash-only sales at \$91 to one-period credit at \$94 with sales rising from 3,850 to 3,940 units. Problem 18: the price per unit under the new policy that makes the switch's NPV zero when new-policy sales are 4,100 units.

Approach. Ross's NPV of switching credit policy (§20.3): the switch costs this period's cash receipts under the old policy plus the variable cost of the extra units, $PQ + v(Q' - Q)$ [20.5]; from next period on, the firm gains the difference in monthly margins forever. Ross's formula [20.6] assumes the same price; with a new price P' the perpetual increment is $(P' - v)Q' - (P - v)Q$. The break-even price sets the NPV to zero.

Notation.

- $P = 91$, P': old and new price; $v = 47$: unit cost (\$); $Q = 3{,}850$, Q': old and new units per period.
- $R = 0.025$: required return per period.

Step 1 — Part 0 — cost of switching.

$$\begin{aligned} \text{Cost} &= PQ + v(Q' - Q) \\ &= (91)(3{,}850) + (47)(3{,}940 - 3{,}850) \\ &= 350{,}350 + 4{,}230 = 354{,}580. \end{aligned} \tag{38}$$

Step 2 — Part 0 — perpetual benefit and NPV.

$$\begin{aligned}\Delta CF &= (P'-v)Q' - (P-v)Q \\ &= (94-47)(3{,}940) - (91-47)(3{,}850) \\ &= 185{,}180 - 169{,}400 = 15{,}780 \text{ per period,} \\ \text{NPV} &= -354{,}580 + 15{,}780/0.025 = -354{,}580 + 631{,}200 = 276{,}620. \end{aligned} \tag{39}$$

Part 0 answer: switch (NPV = \$276,620).

Step 3 — Problem 18 — cost of switching at 4,100 units.

$$\text{Cost} = (91)(3{,}850) + (47)(4{,}100 - 3{,}850) = 350{,}350 + 11{,}750 = 362{,}100.$$

Step 4 — Problem 18 — solve NPV $= 0$ for P'.

$$\begin{aligned} 0 &= -362{,}100 + \frac{(P'-47)(4{,}100) - 169{,}400}{0.025} \\ (P'-47)(4{,}100) - 169{,}400 &= (0.025)(362{,}100) = 9{,}052.5 \\ (P'-47)(4{,}100) &= 169{,}400 + 9{,}052.5 = 178{,}452.5 \\ P' - 47 &= 178{,}452.5/4{,}100 = 43.525 \\ P' &= 47 + 43.525 = 90.525. \end{aligned} \tag{40}$$

Answer. Break-even credit price $P' \approx$ \$90.53 per unit (and in Part 0 the switch has NPV \$276,620).

Intuition. At 4,100 units the extra volume alone pays for one month's delayed receipts, so the firm could even charge slightly *less* on credit (\$90.53) than for cash (\$91) and break even. The credit "markup" needed is negative: the volume effect does the work.

Common mistakes.

- Using the new price in the switching cost: what is lost this period is the old cash revenue PQ, not $P'Q'$.
- Applying Ross's [20.6] with $(P-v)(Q'-Q)$ when the price changes; that formula holds only for $P' = P$.
- Forgetting the extra variable cost $v(Q'-Q)$ in the switching cost.

Ref: Ross Cap. 20, §20.3 "Análise da política de crédito", pp. 712–714 (Eqs. 20.2–20.7). No printed answer.

15. Toys-4-Kids: seasonal vs. level production — Higgins 3.14 (with 3.13), pp. 107–109

What is asked. (a) Redo the quarterly pro forma with level production (COGS 65% of sales, production = annual COGS/4, payables = 10% of average quarterly sales). (b) The effect on annual profit. (c) The effect on quarter-end inventory and on external financing. (d) Whether the firm can borrow that much, the obsolescence risk of building inventory ahead of sales, and whether it worries lenders.

Approach. Percent-of-sales pro forma with external financing required (EFR) as the plug (Higgins Ch. 3, Problem 13): with cash at its \$200 minimum, EFR = total assets − (liabilities + equity); if EFR is negative, set it to zero and let cash absorb the excess. The forecasts hold only on each quarter-end date, which is why seasonality matters (Higgins, "Seasonality", pp. 86–87).

Notation.

- S_q: net sales; $C_q = 0.65S_q$: cost of sales; $Prod = \sum_q C_q/4$: quarterly production; INV_q: end-of-quarter inventory ($ thousand).
- $\tau = 0.40$: tax rate (from the book's table, 188/470); EFR_q: external financing required.

Assumption. As in Problem 13: operating expenses 560 per quarter, receivables 75% of quarterly sales, accrued taxes equal to the quarter's tax (paid in arrears, negative = refund due), net plant 1,000, long-term debt 400, opening equity 3,000, opening inventory 500, minimum cash 200, no dividends; interest on new financing ignored, as the problem instructs. Seasonal figures are recomputed unrounded (the book's table rounds to thousands).

Step 1 — Base case (Problem 13) logic. In Q1, with cash at 200, assets would be 200+225+500+ 1,000 = 1,925 against liabilities and equity of 2,960, so EFR would be −1,035; it is set to 0 and cash becomes 2,960 − 1,725 = 1,235 (Q2 likewise, EFR −727). In Q3 and Q4 EFR is positive: 530.5 and 920.5 (the book prints 530 and 920). Annual profit after tax: $(1-0.40)(0.30\times 8{,}875 - 4\times 560) = 253.5$.

Step 2 — (a) Level production: cost of sales and production. Annual sales $\sum S_q = 300 + 375 + 3{,}200 + 5{,}000 = 8{,}875$.

$$C_q = 0.65S_q = 195,\ 243.75,\ 2{,}080,\ 3{,}250; \quad \textstyle\sum_q C_q = 5{,}768.75,$$
$$Prod = (5{,}768.75)/4 = 1{,}442.1875 \text{ per quarter.}$$

Step 3 — (a) Inventory. $INV_q = INV_{q-1} + Prod - C_q$, starting from 500:

$$\begin{aligned} INV_1 &= 500 + 1{,}442.1875 - 195 = 1{,}747.1875,\\ INV_2 &= 1{,}747.1875 + 1{,}442.1875 - 243.75 = 2{,}945.625,\\ INV_3 &= 2{,}945.625 + 1{,}442.1875 - 2{,}080 = 2{,}307.8125,\\ INV_4 &= 2{,}307.8125 + 1{,}442.1875 - 3{,}250 = 500. \end{aligned}$$

Step 4 — (a) Income statement. Gross profit $S_q - C_q = 105,\ 131.25,\ 1{,}120,\ 1{,}750$; minus 560 gives profit before tax $-455,\ -428.75,\ 560,\ 1{,}190$; tax at 40%: $-182,\ -171.5,\ 224,\ 476$; profit after tax $-273,\ -257.25,\ 336,\ 714$. Equity: $3{,}000 - 273 = 2{,}727$; $2{,}727 - 257.25 = 2{,}469.75$; $+336 = 2{,}805.75$; $+714 = 3{,}519.75$.

Step 5 — (a) Balance sheet and EFR. Payables $= 0.10 \times (8{,}875/4) = 0.10 \times 2{,}218.75 = 221.875$ every quarter. With cash at 200, for Q1: assets $200 + 225 + 1{,}747.1875 + 1{,}000 = 3{,}172.1875$; liabilities and equity $221.875 - 182 + 400 + 2{,}727 = 3{,}166.875$; $EFR_1 = 3{,}172.1875 - 3{,}166.875 = 5.3125$. The other quarters follow in the same way:

Level production ($ thousand)	Qtr 1	Qtr 2	Qtr 3	Qtr 4
Net sales	300.0	375.0	3,200.0	5,000.0
Cost of sales (65%)	195.0	243.8	2,080.0	3,250.0
Gross profit	105.0	131.3	1,120.0	1,750.0
Operating expenses	560.0	560.0	560.0	560.0
Profit before tax	−455.0	−428.8	560.0	1,190.0
Income taxes (40%)	−182.0	−171.5	224.0	476.0
Profit after tax	−273.0	−257.3	336.0	714.0
Cash	200.0	200.0	200.0	200.0
Accounts receivable (75%)	225.0	281.3	2,400.0	3,750.0
Inventory	1,747.2	2,945.6	2,307.8	500.0
Net plant & equipment	1,000.0	1,000.0	1,000.0	1,000.0
Total assets	3,172.2	4,426.9	5,907.8	5,450.0
Accounts payable	221.9	221.9	221.9	221.9
Accrued taxes	−182.0	−171.5	224.0	476.0
Current liabilities	39.9	50.4	445.9	697.9
Long-term debt	400.0	400.0	400.0	400.0
Equity	2,727.0	2,469.8	2,805.8	3,519.8
Total liabilities and equity	3,166.9	2,920.1	3,651.6	4,617.6
External financing required	5.3	1,506.8	2,256.2	832.4

Answer (a). EFR under level production: about $5 thousand, $1,507 thousand, $2,256 thousand and $832 thousand in Q1–Q4 (table above).

Step 6 — (b) Annual profit. Only the cost-of-sales ratio changes, so

$$\Delta PBT = (0.70 - 0.65)(8{,}875) = 443.75,$$
$$\Delta PAT = (1 - 0.40)(443.75) = 266.25 \quad (519.75 \text{ vs. } 253.5).$$

Answer (b). Profit after tax rises by $266.25 thousand, from $253.5 to $519.75 thousand (more than double).

Step 7 — (c) Inventory and financing. Inventory rises above the seasonal 500 by 1,247.2, 2,445.6 and 1,807.8 in Q1–Q3 and is back to 500 in Q4. EFR changes from (0, 0, 530.5, 920.5) to (5.3, 1,506.8, 2,256.2, 832.4), i.e. by +5.3, +1,506.8, +1,725.7 and −88.1. In Q4 less is needed, because accumulated extra profit (+266.25 of equity) and higher accrued taxes (+100) outweigh lower payables (221.875 vs. 500). The average extra quarter-end financing is $(5.3125 + 1{,}506.75 + 1{,}725.6875 - 88.125)/4 = 3{,}149.625/4 = 787.4$; the pre-tax saving of 443.75 would be wiped out only at an interest rate of $443.75/787.4 = 56.36\%$ a year — a rough bound, since it uses quarter-end balances only.

Answer (c). Ending inventory balloons in Q1–Q3 (peak $2.95 million in Q2) and external financing is needed from Q1, peaking at about $2.26 million in Q3 instead of $0.92 million in Q4; by year-end the need is slightly lower ($832 vs. $920 thousand).

Step 8 — (d) Can it borrow, and the obsolescence risk.

- *Capacity.* The peak loan ($2.26 million in Q3) is about $2{,}256/4{,}708 = 47.9\%$ of receivables plus inventory ($2{,}400 + 2{,}307.8 = 4{,}707.8$); in Q2 the loan is 46.7% of $281.3 + 2{,}945.6 = 3{,}226.9$, but there the collateral is almost all inventory. A seasonal, self-liquidating bank line is plausible, but on tougher terms than the base case, where loans were under a quarter of receivables (Higgins's answer to 13(d)).
- *Obsolescence.* Toys are fashion goods: a line built in Q1–Q2 for Christmas may be out of style, mis-forecast or superseded by the time it sells. Unsold goods must be discounted or written off, and the planned 5-point margin gain disappears. Level production converts a demand forecast into a physical bet made months in advance. (Higgins's aside about left shoes is a reminder that what is cheapest to produce is not always what sells.)

- *Lenders.* Yes. Inventory is weak collateral: specific to the firm, valued at cost, and worth little in liquidation if it is last season's toys. A lender will apply a heavy haircut, ask for covenants and monitor sales; repayment depends on the Q3–Q4 selling season actually happening. Receivables in Q3–Q4 are better collateral than inventory in Q1–Q2.

Answer (d). Probably yes, as a seasonal line secured by receivables and inventory, but the lender bears the obsolescence risk of a Q2 balance sheet that is mostly toys not yet sold, and will price and limit the loan accordingly.

Intuition. Level production trades a lower production cost for a larger and earlier investment in inventory. The profit gain is real and large relative to any plausible interest cost; the hidden cost is risk — forecast and obsolescence risk — which falls on the firm and its lenders.

Common mistakes.

- Keeping payables at 10% of each quarter's sales; under level production purchases are smooth, hence payables on average sales.
- Computing inventory as a percentage of sales instead of with the boq/eoq recursion.
- Reading the Q4 EFR (lower than seasonal) as "level production needs less financing"; the peak need is in Q2–Q3.

Book answer. Problem 14 has no printed answer. For Part 0 (Problem 13), Higgins p. 408: negative taxes are a reduction of tax liability (or a rebate); cash exceeds the minimum because EFR would be negative and cash becomes the balancing item; EFR is the balancing item when positive; the firm should easily borrow, as amounts are less than a quarter of receivables. Step 1 matches this.

Ref: Higgins Ch. 3, "Pro Forma Statements" and "Seasonality", pp. 86–87; Problems 13–14, pp. 107–109; answers p. 408.